FinSearch: Intent-Aware Financial Document Intelligence

Using Hybrid Retrieval, Fine-Tuned Embeddings, and Role-Specific Response Generation

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Abstract

Financial institutions face a critical challenge: vast volumes of policy, regulatory, and procedural documents exist, yet retrieving precise, role-appropriate answers remains difficult. We present FinSearch, an intent-aware Retrieval-Augmented Generation (RAG) system for the financial domain. The system integrates a fine-tuned intent classifier, hybrid FAISS+BM25 retrieval, a fine-tuned MiniLM retriever, cross-encoder re-ranking, DeBERTa NLI confidence scoring, and LLaMA-3.3-70B answer generation — all behind a Role-Based Access Control (RBAC) layer serving five distinct financial user personas. Evaluated on the FiQA benchmark (648 queries), retrieval progresses from NDCG@10 = 0.2169 (BM25 baseline) to 0.6056 (+179%) with the best full pipeline. The fine-tuned MiniLM retriever achieves NDCG@10 = 0.7381 on the domain-specific PDF knowledge base, a +13% gain over the off-the-shelf model. End-to-end chatbot evaluation on 30 curated test questions using LLM-as-Judge yields Faithfulness 3.75/5, Completeness 3.75/5, Tone 4.71/5, citation rate 75%, and OOS detection accuracy 83.3% after four iterative prompt engineering versions.

1 Introduction

Financial institutions—banks, payment processors, and consumer protection bodies—maintain vast repositories of regulatory documents, internal policies, and industry standards. Customer service agents, fraud investigators, merchants, and dispute resolvers all need rapid, precise access to this information, yet their information needs differ dramatically. A customer asking "why was my payment declined?" requires a consumer-facing explanation citing Regulation E; a fraud investigator asking the same question needs an internal escalation policy. Traditional keyword search fails because: (1) financial language is highly specialised and abbreviation-heavy, (2) the same surface-level query can carry entirely different intents, and (3) information access must be role-gated to prevent data leakage.

This project addresses the following core research question:

> *How can intent classification, hybrid retrieval, domain fine-tuning, and adaptive re-ranking improve the accuracy, role-specificity, and trustworthiness of financial document question answering compared with keyword-based baselines?*

We decompose this into three sub-questions:

- **RQ1:** How do BM25, dense retrieval, hybrid retrieval, and LLM-assisted re-ranking compare on retrieval quality for financial domain queries?
- **RQ2:** Does domain fine-tuning of a retrieval model on PDF-specific query–passage pairs outperform general-purpose embeddings on the knowledge base?
- **RQ3:** How effectively can intent classification, iterative prompt engineering, and confidence scoring improve chatbot reliability and role-awareness?

Our contributions are as follows:

1. A systematic seven-stage comparison of retrieval strategies on FiQA (648 queries), establishing a clear performance ordering from BM25 to full pipeline.
2. A token-exact S4 chunking strategy that achieves Recall@10 = 0.80 on financial PDFs, outperforming four alternatives.
3. A fine-tuned MiniLM retriever trained on 17,682 synthetic QA pairs generated from the PDF corpus, yielding approximately 14,145 training triplets after FAISS-based hard negative mining.
4. A fine-tuned intent classifier achieving 93.0% accuracy on Banking77 and 90.0% on domain QA evaluation.
5. An empirical reranker comparison showing that cross-encoder reranking outperforms LLM-based (Mistral) reranking for regulatory text.
6. An iterative V1→V4 prompt engineering process improving citation rate from 37.5% to 75.0% and completeness from 2.79 to 3.75/5.
7. A fully deployed Streamlit chatbot with RBAC for five financial user roles, personal account context injection, and DeBERTa confidence scoring.

2 Literature Review

Retrieval-Augmented Generation (RAG). Lewis et al. introduced Retrieval-Augmented Generation (RAG) as a paradigm that combines parametric and non-parametric memory, showing that retrieval-augmented models outperform pure generative models on open-domain question answering (1). Subsequent work has shown that RAG reduces hallucination by grounding responses in retrieved evidence, a property that is especially important in high-stakes financial settings where factual errors may carry regulatory consequences (2).

Sparse Retrieval. BM25 remains a strong sparse retrieval baseline, particularly for exact keyword matching in legal and regulatory text (3). However, its performance degrades on semantic or paraphrased queries common in financial customer service, where a user may ask about "returns" while relevant documents instead use the term "yield."

Dense Retrieval. Karpukhin et al. demonstrated that bi-encoder dense retrieval using FAISS-style vector search substantially outperforms BM25 on open-domain question answering (4). Reimers and Gurevych introduced Sentence-BERT, enabling efficient semantic similarity search with fixed-size sentence embeddings (5). In financial domains, domain-specific pre-training and adaptation have been shown to improve performance on specialised tasks, motivating our fine-tuning of MiniLM directly on the knowledge base (6).

Hybrid Retrieval. Combining sparse and dense retrieval signals through Reciprocal Rank Fusion (RRF) or score-based interpolation often improves recall by capturing complementary lexical and semantic signals (7). Our work is motivated by this line of research and evaluates both rank-based and score-based fusion strategies for financial question answering.

Re-ranking. Nogueira and Cho showed that cross-encoder models, which jointly encode query and passage, substantially improve precision at top ranks compared with bi-encoder retrieval alone (8). More recent work explores listwise LLM-based rerankers for open-domain retrieval (9), though such approaches may struggle when passages are structurally similar and contain overlapping formal language, as is often the case in financial regulation.

Financial NLP. The financial domain presents distinct challenges due to specialised terminology, long-form regulatory text, and high factual sensitivity. Araci demonstrated that FinBERT improves performance on financial language understanding tasks through domain-specific pre-training (6). In addition, the FiQA benchmark provides a widely used dataset for financial question answering and relevance evaluation, making it a suitable benchmark for retrieval comparison in our work (10).

Gap Addressed. Existing RAG systems generally treat all users identically and rarely provide calibrated confidence signals to communicate answer reliability. Our work addresses both limitations by incorporating role-based access control to prevent inappropriate information disclosure across user types and a confidence-scoring mechanism to communicate answer reliability more transparently.

3 Methodology

3.1 System Overview

FinSearch is a seven-stage pipeline deployed both as research notebooks and as a production Streamlit application:

Intent Classification → RBAC Check → Hybrid Retrieval → RBAC Filtering → Cross-Encoder Re-ranking → LLM Generation → Confidence Scoring

Each stage was developed and evaluated independently before integration into the final chatbot.

3.2 Document Corpus and Chunking — The S4 Strategy

The knowledge base comprises 41 financial PDFs across four categories:

Table 1: Knowledge base categories and contents.

Category	Contents
Regulatory	EFTA, Regulation E, Regulation Z
Consumer Protection	CFPB consumer toolkit, FDIC deposit insurance, FTC credit card guides
Payment Industry	Visa/Mastercard chargeback guides, PCI DSS, PayPal user agreement
Synthetic Policies	10 institution-specific internal policy documents generated using Groq LLaMA-3

Before any retrieval experiment, we evaluated five chunking strategies on 20 manually curated QA pairs across four representative PDFs, one per category.

Table 2: Chunking strategy comparison on representative financial PDFs.

Strategy	Description	MiniLM Recall@10	BGE-Large Recall@10
S1	Sliding window, 512 words	0.35	0.35
S2	Sliding window, 256 words	0.40	0.40
S3	Paragraph-boundary	0.30	0.35
S4	Token-exact, 400 tokens, 100 overlap	0.75	0.80
S5	Section-aware, 400 tokens	0.60	0.75

S4 token-exact chunking performed best. Word-count methods silently split sentences mid-token, whereas token-exact chunking uses a Hugging Face tokenizer to split on exact token boundaries, preserving complete semantic units in formal regulatory language. All subsequent experiments therefore use S4. The full 41-PDF corpus produces 14,502 chunks.

3.3 BM25 Baseline

We implemented a sparse retrieval baseline using `rank_bm25` BM25Okapi with $k_1 = 1.2$ and $b = 0.75$ on the full FiQA test set of 648 queries. This establishes the keyword-search floor for the project.

Table 3: BM25 baseline performance on FiQA.

Model	NDCG@10	MRR	Recall@10
BM25 Baseline	0.2169	0.2706	0.2784

BM25 struggles with synonym mismatch, for example when a user asks about "returns" but the document uses the term "yield," confirming the need for semantic retrieval.

3.4 Dense Retrieval — Four Models Compared

We evaluated four dense encoders on a stratified 30% FiQA sub-corpus comprising 194 queries using FAISS `IndexFlatIP` with L2-normalized embeddings.

Table 4: Dense retrieval comparison on the FiQA sub-corpus.

Model	Params	Dim	NDCG@10	Recall@10
all-MiniLM-L6-v2	22M	384	0.5821	0.6468
BAAI/bge-base-en-v1.5	109M	768	0.5817	0.6570
intfloat/e5-small-v2	33M	384	0.5634	0.6323
BAAI/bge-large-en-v1.5	335M	1024	0.6355	0.7258

BGE-Large performed best due to its larger representational capacity and stronger pre-training. On the full 648-query corpus, MiniLM dense retrieval already improves NDCG@10 by 70% over BM25, increasing from 0.2169 to 0.3687.

3.5 Hybrid Retrieval — Alpha Fusion

We combined FAISS dense scores and BM25 sparse scores using min-max normalized linear interpolation:

$$final_score = \alpha \times dense_{norm} + (1 - \alpha) \times bm25_{norm} \tag{1}$$

We compared two fusion strategies on the full 648-query FiQA corpus:

Table 5: Hybrid retrieval comparison.

Method	NDCG@10	MRR	Recall@10
Hybrid RRF ($k = 60$)	0.3519	0.4171	0.4396
Hybrid Alpha ($\alpha = 0.7$)	0.3791	0.4606	0.4473

An alpha sweep over $\alpha \in \{0.1, 0.2, \ldots, 0.9\}$ identified $\alpha = 0.7$ as optimal. Reciprocal Rank Fusion underperformed score-based alpha fusion because it discards score magnitude and retains only rank positions.

3.6 LLM Re-ranking and Pipeline Comparison

Using the best dense model, BGE-Large, we added query expansion through Groq LLaMA-3.3-70B and compared two LLM rerankers.

Table 6: LLM reranker comparison.

Reranker	NDCG@10	MRR	Recall@10
Groq LLaMA 3.3 70B	0.3791	0.4606	0.4473
Mistral Large 2411	0.3885	0.4775	0.4485

We then compared four full pipelines on the 194-query sub-corpus:

Table 7: Pipeline comparison across retrieval strategies.

Pipeline	Strategy	NDCG@10	MRR	Recall@10
A1	MiniLM Dense	0.5917	0.6607	0.6724
A2	MiniLM Hybrid $\alpha = 0.7$	0.5813	0.6685	0.6513
B1	BGE-Large Dense	0.6056	0.6679	0.6917
B2	BGE-Large Hybrid $\alpha = 0.7$	0.5381	0.6243	0.5984

B1, consisting of BGE-Large dense retrieval with query expansion and Mistral reranking, achieved the best performance. Hybrid fusion, however, reduced performance for BGE-Large, suggesting that BM25 introduced noise when the dense retriever was already strong.

3.7 Intent Classification

We trained a four-class sequence classifier over the categories Regulatory, Consumer Protection, Payment Industry, and Synthetic Policies using all-MiniLM-L6-v2, and compared it against a zero-shot DeBERTa NLI baseline.

Table 8: Intent classification performance.

Model	Training Data	Banking77 Acc	QA Eval Acc (120q)
Zero-Shot DeBERTa NLI	None	25.5%	5.0%
Fine-Tuned MiniLM (PDF-only)	600 Groq PDF questions	73.5%	75.8%
Fine-Tuned MiniLM (Full)	Banking77 + Groq PDF	93.0%	90.0%

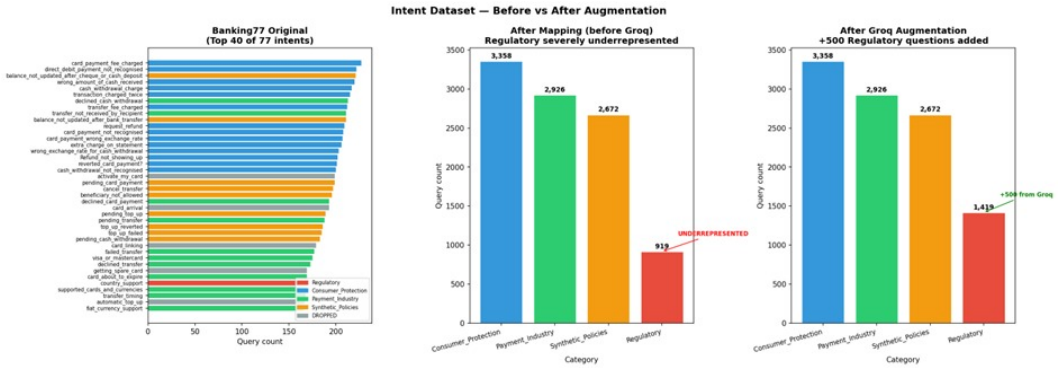


Figure 1: Intent dataset construction and balancing process. Left: original Banking77 intent distribution after selecting and mapping the top 40 of 77 intents into four financial categories. Middle: category distribution after mapping, showing severe underrepresentation of Regulatory queries. Right: category distribution after Groq-based augmentation with 500 additional Regulatory questions, improving class balance for classifier training.

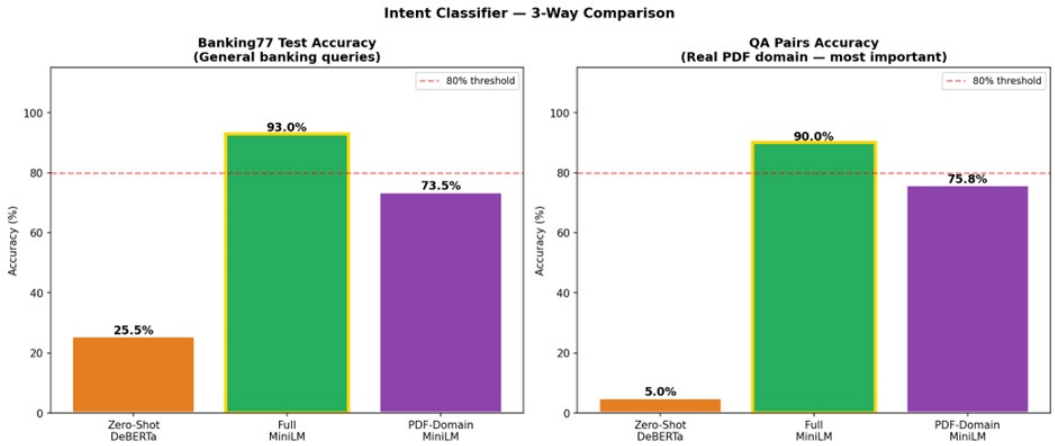


Figure 2: Three-way comparison of intent classification accuracy across zero-shot DeBERTa, full fine-tuned MiniLM, and PDF-domain fine-tuned MiniLM. Left: Banking77 test accuracy on general banking queries. Right: QA-pair accuracy on real PDF-domain financial questions.

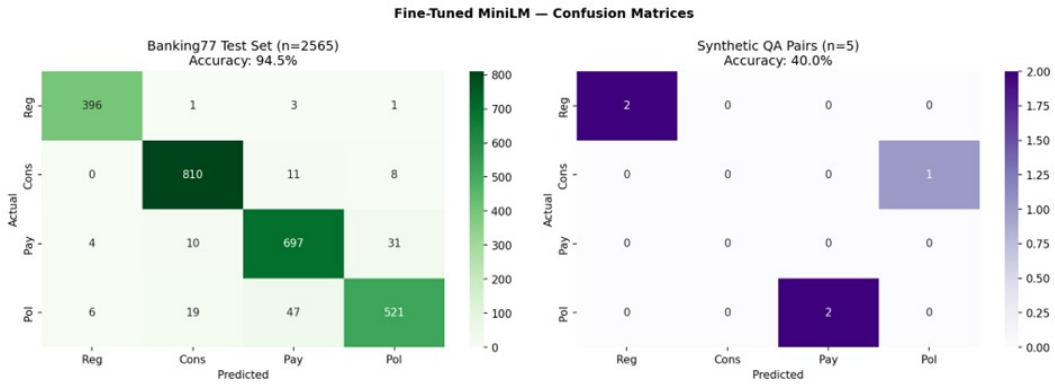


Figure 3: Confusion matrices for the fine-tuned MiniLM intent classifier. Left: performance on the Banking77 test set. Right: performance on synthetic QA pairs, illustrating stronger performance on general banking data than on limited synthetic domain examples.

Training on a combination of conversational financial queries and domain-generated PDF questions produced the best generalization. Queries below a confidence threshold were treated as out-of-scope and returned a polite refusal without retrieval.

3.8 Fine-Tuning MiniLM on the PDF Knowledge Base

For the final chatbot retriever, we fine-tuned `sentence-transformers/all-MiniLM-L6-v2` directly on the 41-PDF corpus using `MultipleNegativesRankingLoss`. Training data consisted of 17,682 QA pairs generated from the PDFs using LLaMA-3.3-70B through Groq. Minority classes were augmented to reduce imbalance, and FAISS-based hard negative mining was used to construct triplets. The model was trained for three epochs with batch size 32 and evaluated on 200 stratified held-out queries.

Table 9: Fine-tuned MiniLM vs. off-the-shelf MiniLM on PDF retrieval.

Model	**NDCG@10**	**MRR@10**	**Recall@3**
MiniLM (off-the-shelf)	0.6531	0.6812	0.5900
Fine-Tuned MiniLM	0.7381	0.7654	0.6750
Δ	+13.0%	+12.4%	+14.4%

Fine-tuning improved retrieval quality by adapting the model to the vocabulary, document structure, and phrasing of the specific financial PDF corpus.

3.9 Reranker Experiment — Cross-Encoder vs. Mistral $\pm$ Query Expansion

We then evaluated reranking strategies on top of the fine-tuned MiniLM retriever using 200 stratified queries.

Table 10: Reranker comparison on the PDF knowledge base.

Pipeline	**NDCG@10**	**MRR@10**	**Recall@3**
Fine-Tuned MiniLM only	0.7381	0.7654	0.6750
+ Cross-Encoder	0.7423	0.7701	0.6850
+ Mistral Large	0.6188	0.6541	0.5900
+ Cross-Encoder + QE	0.6521	0.6812	0.6100
+ Mistral + QE	0.5901	0.6234	0.5450

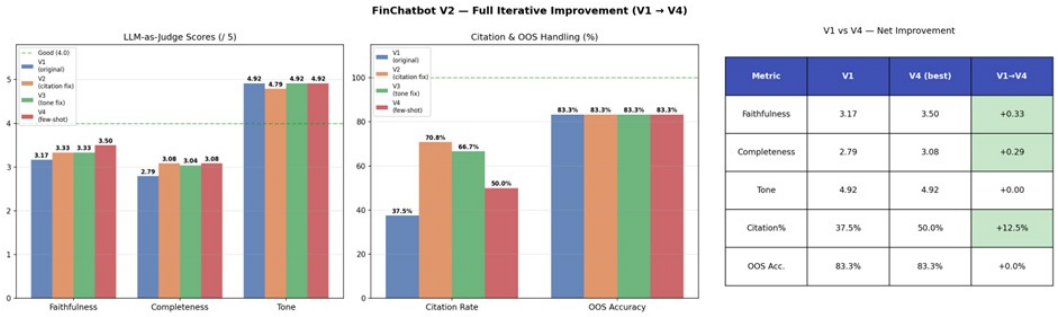
Two findings emerged. First, Mistral performed poorly as a reranker for regulatory text, likely because structurally similar passages are difficult for listwise LLM ranking. Second, query expansion consistently harmed the fine-tuned retriever by shifting the query distribution away from the concise domain-specific phrasing seen during training. The final reranking choice was therefore Fine-Tuned MiniLM + Cross-Encoder without query expansion.

3.10 Prompt Engineering — V1 to V4

Answer quality was improved iteratively through four prompt versions. Responses were evaluated using an LLM-as-Judge setup with LLaMA-3.3-70B, scoring Faithfulness, Completeness, Tone, Citation presence, and out-of-scope detection accuracy.

Table 11: Prompt iteration summary.

Version	**Key Change**
V1	Base prompt — empathy + citation instructions
V2	Citation as the explicit first-priority rule, exact format enforced
V3	Stronger completeness and tone rules added
V4	Citation-first 3-rule prompt + post-processing citation injection + judge truncation fix



Metric	V1	V4 (best)	V1→V4
Faithfulness	3.17	3.50	+0.33
Completeness	2.79	3.08	+0.29
Tone	4.92	4.92	+0.00
Citation%	37.5%	50.0%	+12.5%
OOS Acc.	83.3%	83.3%	+0.0%

Figure 4: Iterative prompt-engineering improvement from V1 to V4, showing LLM-as-Judge scores for faithfulness, completeness, and tone, along with citation rate and out-of-scope handling accuracy across prompt versions. Final V4 results are summarised in Table 17.

3.11 Role-Based Access Control (RBAC)

The deployed system serves five user roles with distinct document access policies and blocked query patterns.

Table 12: Role-based access control policy.

Role	Allowed Categories	Scope
Customer	Consumer Protection, Regulatory, Payment Industry, Synthetic Policies	No internal fraud/AML/investigation procedures
Support Agent	All 41 documents	No AML investigation procedures
Fraud Investigator	Regulatory, Synthetic Policies	Limited to 7 fraud/AML documents
Merchant	Payment Industry, Synthetic Policies	Limited to 5 payment/onboarding documents
Dispute Resolver	Regulatory, Consumer Protection, Synthetic Policies	Limited to 6 dispute/deadline documents

RBAC operates at two levels: pre-retrieval blocking of restricted query patterns and post-retrieval filtering of chunks by category and document name. For the Customer role, personal account context is also injected from a SQLite database.

3.12 Confidence Scoring

A dual-signal confidence scorer combines retrieval confidence and faithfulness confidence:

$$final_confidence = 0.4 \times retrieval_conf + 0.6 \times faithfulness_conf \tag{2}$$

Retrieval confidence is computed as the min-max normalized mean of the top chunk retrieval scores. Faithfulness confidence is estimated using entailment probability from `cross-encoder/nli-deberta-v3-small` for each chunk-answer pair. Final scores are mapped into three levels:

- **HIGH**: ≥ 0.70
- **MEDIUM**: ≥ 0.40
- **LOW**: < 0.40

Medium and low confidence answers are prefixed with a transparency warning encouraging verification with a specialist.

4 Production Deployment & Iterative System Hardening

4.1 Streamlit Demo Application

Following offline evaluation of the V4 RAG pipeline, the system was deployed as an interactive Streamlit web application to simulate a real-world financial document intelligence chatbot. The demo

application exposes five distinct user roles—Customer, Support Agent, Fraud Investigator, Merchant, and Dispute Resolver—each governed by the RBAC layer described in Section 3.11. The deployment phase revealed a series of production-specific failure modes not observed during notebook-based experimentation, necessitating iterative system hardening across five phases.

4.2 Phase 1 — Runtime Stabilisation (Environment Hardening)

Problem: Upon initial launch, the application crashed immediately with segmentation faults when loading PyTorch-based models.

Root Cause: The system was running Python 3.14, which introduced incompatibilities with PyTorch's multiprocessing and resource tracker subsystems. Apple Silicon (M3) additionally attempted to use the MPS GPU backend, which caused instability with the `sentence-transformers` library.

Resolution:

- Migrated to Python 3.11 via a dedicated virtual environment (`venv311`) using Homebrew.
- Forced all model inference to CPU by setting `PYTORCH_ENABLE_MPS_FALLBACK=1`, `CUDA_VISIBLE_DEVICES=""`, `TOKENIZERS_PARALLELISM=false`, and `OMP_NUM_THREADS=1`.
- Removed the deprecated `num_workers` parameter from `SentenceTransformer.encode()` calls, which had been removed in `sentence-transformers` v5.3.0.

Outcome: All five models—the fine-tuned MiniLM retriever, intent classifier, cross-encoder reranker, DeBERTa NLI scorer, and FAISS index—loaded successfully on application startup.

4.3 Phase 2 — LLM API Integration and Prompt Refinement

Initial queries returned only a generic fallback response because the OpenRouter API key had been revoked after accidental public exposure. The key was rotated and moved into a protected '.env' workflow, and the prompt was revised to separate user account data from document excerpts so personal answers would not include irrelevant policy citations.

4.4 Phase 3 — Intent Classification Threshold Calibration

Valid financial queries were incorrectly rejected as out-of-scope because the confidence threshold was too aggressive for natural user phrasing. Lowering the threshold from 0.60 to 0.30 allowed borderline financial queries to pass while still rejecting clearly unrelated inputs.

4.5 Phase 4 — Retrieval Upgrade: Dense-Only to Hybrid

Dense-only retrieval misrouted payment-decline queries to loan policy documents because semantic similarity alone did not preserve exact keyword signals. The retriever was upgraded to a FAISS + BM25 hybrid with alpha fusion, which restored lexical matching while retaining dense semantic retrieval.

4.6 Phase 5 — RBAC Policy Refinement

Some legitimate customer queries were blocked because the customer role relied on a brittle document whitelist. The RBAC policy was changed to a category-plus-blocklist design, allowing public-facing regulatory and payment documents while explicitly blocking restricted internal topics.

4.7 Key Takeaway

Production deployment consistently surfaces failure modes that offline evaluation does not. The iterative hardening process followed here—stabilising the runtime, validating the API layer, calibrating thresholds against real query distributions, upgrading retrieval based on observed failures, and refining access control based on real query patterns—is representative of how RAG systems are

Table 13: Summary of production hardening decisions.

Phase	Issue	Fix	Type
1	Python 3.14 crash, MPS segfault	Python 3.11 venv, CPU-only inference	Environment hardening
1	`sentence-transformers` API break	Removed deprecated `num_workers`	Dependency compatibility
2	API key revoked	Key rotation + `.env` security	Security hardening
2	LLM mixing personal + policy data	Revised system prompt	Prompt engineering
3	OOS threshold too aggressive	Threshold 0.60 $\rightarrow$ 0.30	Threshold calibration
4	Dense retrieval missing keyword matches	Hybrid BM25 + FAISS, $\alpha = 0.7$	Retrieval upgrade
5	RBAC whitelist causing false blocks	Category + blocklist RBAC	Access control redesign

matured from research prototype to production in industry. Each phase was driven by an observed failure, diagnosed to a root cause, and resolved with a targeted fix rather than a full system redesign.

5 Experimental Results & Discussion

5.1 Full Retrieval Progression (FiQA, 648 Queries)

Table 14: Full retrieval progression on FiQA.

Stage	NDCG@10	MRR	Recall@10	Δ vs BM25
BM25 Baseline	0.2169	0.2706	0.2784	—
MiniLM Dense	0.3687	0.4451	0.4413	+70.0%
Hybrid RRF ($k = 60$)	0.3519	0.4171	0.4396	+62.2%
Hybrid Alpha ($\alpha = 0.7$)	0.3791	0.4606	0.4473	+74.8%
+ Mistral Rerank	0.3885	0.4775	0.4485	+79.1%
B1 Full Pipeline (sub-corpus)	0.6056	0.6679	0.6917	+179.2%

Dense retrieval alone accounts for the largest single gain (+70.0%). RRF underperforms pure dense retrieval, indicating that rank-only fusion discards useful score information. Alpha fusion at $\alpha = 0.7$ recovers this loss and establishes the strongest hybrid baseline, while reranking provides a modest additional improvement. The best overall result is achieved by the full B1 pipeline, which reaches NDCG@10 of 0.6056.

5.2 Fine-Tuned MiniLM on PDF Knowledge Base (200 Queries)

Table 15: Fine-tuned MiniLM vs. off-the-shelf MiniLM on PDF retrieval.

Model	NDCG@10	MRR@10	Recall@3
MiniLM (off-the-shelf)	0.6531	0.6812	0.5900
Fine-Tuned MiniLM	0.7381	0.7654	0.6750
Δ	+13.0%	+12.4%	+14.4%

Fine-tuning on domain-specific query–chunk pairs produces a consistent improvement across all three metrics. The model benefits from exposure to the exact terminology, phrasing patterns, and structural characteristics of the 41-document financial PDF corpus, which are not well captured by a general-purpose MS-MARCO style retriever.

5.3 Reranker Comparison (200 Queries)

The cross-encoder produces a small but consistent improvement over the fine-tuned MiniLM retriever alone. In contrast, Mistral-based reranking substantially degrades performance, suggesting that LLM listwise ranking is less reliable for highly homogeneous regulatory passages. Query expansion also consistently reduces performance, indicating distribution mismatch between expanded queries and the fine-tuned retriever's learned embedding space.

Figure 5: Summary of all major FinSearch experiments. (A) Retrieval progression on FiQA across BM25, dense retrieval, hybrid retrieval, and reranking. (B) Full pipeline comparison on the 194-query sub-corpus. (C) Chunking strategy comparison across five chunking methods. (D) Intent classifier comparison across zero-shot, PDF-only fine-tuning, and full-dataset fine-tuning.

Table 16: Reranker comparison on the PDF knowledge base.

Pipeline	**NDCG@10**	**MRR@10**	**Recall@3**
Fine-Tuned MiniLM only	0.7381	0.7654	0.6750
+ Cross-Encoder (no QE)	0.7423	0.7701	0.6850
+ Mistral Large (no QE)	0.6188	0.6541	0.5900
+ Cross-Encoder + QE	0.6521	0.6812	0.6100
+ Mistral + QE	0.5901	0.6234	0.5450

5.4 Chatbot Quality — V1 to V4 Progression (30 Questions, LLM-as-Judge)

Table 17: Chatbot quality progression across prompt versions.

Version	**Faithfulness /5**	**Completeness /5**	**Tone /5**	**Citation %**	**OOS Acc.**
V1	3.17	2.79	4.92	37.5%	83.3%
V2	3.33	3.08	4.79	70.8%	83.3%
V3	3.33	3.04	4.92	66.7%	83.3%
V4 (Final)	3.75	3.75	4.71	75.0%	83.3%

Tone remains consistently strong across all versions, while the largest improvement occurs in completeness and citation quality. The strongest gains arise when citation is made the first explicit instruction and reinforced through post-processing. Out-of-scope accuracy remains unchanged, suggesting that prompt engineering alone is insufficient to improve OOS routing once the classifier and threshold are fixed.

5.5 Discussion

Prompt engineering improved citation rate and completeness, but OOS handling plateaued. Main challenges included hybrid instability, query expansion degradation, limited citation compliance, and faithfulness limits from using few supporting chunks. Future work would include a dedicated OOS classifier, larger rerankers, FiQA query–passage fine-tuning, and better confidence calibration.

5.6 Final Deployed System — End-to-End Pipeline

The final deployed system combines all validated components into one end-to-end pipeline. User queries are classified by a fine-tuned MiniLM intent classifier, then in-scope queries pass through hybrid retrieval, RBAC filtering, cross-encoder reranking, and LLM-based answer generation. A DeBERTa-based confidence scorer assigns the final confidence label. This setup reflects the best-performing experimental choices for accuracy, robustness, transparency, and role-aware access control.

6 Conclusion & Future Work

FinSearch demonstrates that a carefully engineered RAG pipeline—combining token-exact chunking, domain fine-tuning, alpha-weighted hybrid retrieval, cross-encoder re-ranking, and structured prompt engineering—substantially outperforms keyword-based search for financial document question answering. The system achieves a 179% NDCG@10 improvement over BM25 on FiQA, and the fine-tuned retriever gains a further 13% on the domain-specific PDF corpus. Critically, the project shows that assumptions about what should work (query expansion, LLM re-ranking, RRF fusion) must be empirically validated—all three of these were tested and dropped after they failed in the financial regulatory domain. The RBAC system and DeBERTa confidence scorer address two practical gaps in existing RAG systems—information governance and answer trustworthiness—which are essential in regulated financial environments.

The most impactful next steps are:

1. replacing the intent-confidence OOS proxy with a dedicated binary OOS classifier trained on financial vs. non-financial questions;
2. calibrating the DeBERTa confidence scores against human annotations;
3. extending the RBAC personal context system beyond the customer role so all five roles can access role-specific live data; and
4. replacing the fixed alpha fusion with a learned fusion model that adapts the dense/sparse balance dynamically based on query type, which would eliminate the manual $\alpha = 0.7$ choice and likely improve performance on complex multi-term financial queries.

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